ZPDPatch: Separating Trajectory Supervision from Checkpoint Diversity in Execution-Guided Program Repair

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Automated program repair for programming assignments usually treats the latest submission as an isolated buggy program. We present ZPDPatch, which mines execution-ordered Progress, Strict, and terminal Answer relations from authentic submission trajectories, trains independent checkpoints, and selects a size-three execution portfolio. Its controller guarantees observed-test non-regression and optional AST edit-budget compliance. We evaluate Qwen2.5-Coder-7B adapters on 997 Seen and 250 problem-held-out trajectories, plus separately trained student-held-out Java models. On Seen, ZPDPatch raises repair rate from 30.7% for trajectory-prompted Zero-shot to 61.5%. A matched $A_1 \rightarrow A_3 \rightarrow A_9 \rightarrow M_9$ ladder shows that independent checkpoint diversity explains the dominant unrestricted gain: $A_3 - A_1 = +11.63$ points [9.54, 13.74], while selection-matched mixed-target $M_9 - A_9 = +1.71$ [−0.79, 4.29] is uncertain. Relation targets instead affect patch scope. Across six fixed edit budgets, M_9 gains 1.22 mean Seen points [0.02, 2.42] over A_9 and preserves more buggy tokens and source lines on joint repairs. Problem-disjoint selection reverses their unrestricted ranking but retains mixed-target gains at two edit budgets. Both pools reach 74.8% unrestricted repair on held-out problems; hidden-test and Java contrasts are inconclusive. A target-matched ablation finds no stable benefit from serializing earlier submissions. LSGen repairs more programs without an edit limit, whereas ZPDPatch leads its mean edit-budget frontier by 14.68 points [12.03, 17.43]. Thus checkpoint diversity explains coverage, while trajectory-mined targets change the executable locality frontier rather than establishing a history effect.

CCS Concepts: • **Software and its engineering** → **Software testing and debugging**; • **Applied computing** → *Education*; • **Computing methodologies** → *Natural language generation.*

Additional Key Words and Phrases: automated program repair, programming education, large language models, submission trajectories, execution-guided repair

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1 Introduction

Programming students iteratively write, submit, execute, and revise code. Online judges make this loop scalable, but their feedback is usually limited to coarse verdicts such as `Wrong Answer`, `Runtime Error`, or `Time Limit Exceeded`. A verdict reports failure without explaining whether the latest revision made progress, repeated an earlier defect, or moved away from a viable solution. Instructors cannot inspect every trajectory and author an individualized repair at the scale of modern programming courses.

Automated program repair (APR) offers executable, code-specific feedback. The field spans search-based generate-and-validate repair, learned transformations, and test-guided validation [Gazzola et al. 2019; Le Goues et al. 2012; Monperrus 2018; Weimer et al. 2009]. Systems for programming assignments cluster peer solutions, align a student's code to a reference, or synthesize a fixed program [Gulwani et al. 2018; Hu et al. 2020; Wang et al. 2018]. More recent systems use language models, execution diagnostics, retrieved examples, and iterative conversations [Joshi et al. 2023; Liu et al. 2024; Orvalho et al. 2025; Yang et al. 2024; Zhang et al. 2024]. Most of these approaches condition on the latest buggy program, possibly augmented with tests or peer programs. The earlier submissions produced by the same student are rarely treated as first-class repair context.

The motivation for using history is straightforward but unverified. Two students may arrive at similar current programs through different revisions. Their histories expose repeated errors, attempted edits, and states that they demonstrably reached. In educational terms, such reachable

improvements resemble the Zone of Proximal Development (ZPD): states attainable with appropriate support but not yet reached independently [Chounta et al. 2017; Cole et al. 1978; Murray and Arroyo 2002]. We therefore ask whether repair can exploit a student's observed trajectory to produce a useful next program rather than immediately replacing it with an arbitrary correct solution. We use ZPD as a design motivation, not as a measured psychological construct or a claim about learning outcomes.

We introduce ZPDPatch, illustrated in Figure 1. It automatically converts authentic submission trajectories into three supervised repair datasets. Progress retains verdict improvements and same-verdict transitions whose per-test outcomes improve monotonically. Strict retains only transitions that improve the online-judge verdict. Answer pairs every failed submission with the trajectory's final accepted program. Three seeds per target relation produce nine independently trained QLoRA checkpoints over the same code language model and prompt. Validation execution selects three complementary checkpoints, which are invoked from narrower to broader supervision. ZPDPatch executes each candidate, stops on an eligible acceptance, and otherwise selects a non-regressive candidate using test pass rate and AST tree edit distance (TED).

Our evaluation separates mechanisms that are easy to conflate: repeated calls, independent training seeds, validation-selected pool breadth, relational target composition, and serialization of prior submissions. The $A_1 \rightarrow A_3 \rightarrow A_9 \rightarrow M_9$ ladder matches each successive opportunity. Target-matched retraining separately finds no stable causal benefit from serializing earlier submissions.

This paper makes the following contributions:

- We formulate trajectory-mined repair as finite execution-evidence orders plus terminal closure, and audit every materialized supervision transition against those definitions.
- We pose size-three checkpoint choice as exact finite validation-coverage search, derive a problem-level finite-family selection bound, and design an execution controller with observed-test non-regression and structural edit-budget compliance by construction.
- We empirically separate relation targets from seed and selection diversity using the full A_1–A_3–A_9–M_9 ladder, target-matched history ablations, hidden tests, problem-disjoint selection, patch-budget and source-retention analyses, a second model scale, and student- and exercise-held-out Java replications.

2 Background and Related Work

2.1 Structure- and Reference-Based Educational Repair

Educational APR can exploit many programs written for one specification. CLARA clusters and aligns programs before extracting a repair; SARFGEN searches and aligns reference programs; Refactory normalizes refactorings before comparison [Gulwani et al. 2018; Hu et al. 2020; Wang et al. 2018]. Embeddings, solution-space models, and visualization propagate annotations or expose recurring strategies, while multi-function and diversified methods broaden the target beyond one correction [Choi et al. 2021; Choi and Lee 2025; Glassman et al. 2015; Heo et al. 2023; Piech et al. 2015b; Rivers and Koedinger 2017; Yi et al. 2017]. These methods provide strong assignment-specific evidence and establish that execution correctness and relationship to the student's program are separate repair properties.

LSGen is our reference-rich baseline. It filters historical repair pairs, retrieves by edit information, generates, and may retrieve again from the new state [Dai et al. 2026]. We retain its access to other users' Seen-Train solutions for the target problem. That information is unavailable on genuinely new assignments, so LSGen is not evaluated on problem-held-out Unseen tasks. This separates repository-mature repair from repair using only a learned model, statement, and tests.

2.2 Neural and Large-Language-Model Repair

Neural repair learned transformations from repository histories and context-aware translation before pretrained code models enabled zero-shot and fine-tuned repair [Jiang et al. 2023; Lutellier et al. 2020; Tufano et al. 2019; Xia and Zhang 2022; Yasunaga and Liang 2021]. RING, PyDex, and FastFixer cast educational repair as conditional generation [Joshi et al. 2023; Liu et al. 2024; Zhang et al. 2022, 2024]; retrieval and localization further constrain generation with peer programs or high-precision syntax feedback [Phung et al. 2023; Zhao et al. 2024]. These systems cover programs that do not align cleanly with a reference, but make correctness and patch scope external validation concerns.

Execution-aware methods address those concerns with signals independent of model confidence. SelfAPR learns from test diagnostics; FixEval makes execution the repair-evaluation criterion; CREF carries diagnostics through a repair conversation; counterexample-guided repair combines generation, fault localization, and MaxSAT [Haque et al. 2023; Orvalho et al. 2025; Yang et al. 2024; Ye et al. 2022]. ZPDPatch likewise executes candidates, but introduces a different unit of supervision and control. Authentic trajectories define nested improvement relations rather than synthetic corruptions or manually assigned repair roles. Members share a base model and may have correlated errors, but generation is non-adaptive: no member consumes another member's output. The unchanged program remains selectable, so a failed generation cannot force a regression.

2.3 Adaptive Feedback, ZPD, and Submission Histories

Programming feedback ranges from correctness reports to next-step hints and complete repairs; feedback research emphasizes timeliness, task focus, and actionable information rather than correctness alone [Hattie and Timperley 2007; Keuning et al. 2018; Narciss 2008; Shute 2008]. Tutoring research studies help seeking and assistance at a productive granularity; iSnap and LLM hint systems similarly expose next-step or multi-level support [Aleven and Koedinger 2000; Heickal and Lan 2024; Price et al. 2017; Roest et al. 2024; Xiao et al. 2024]. The ZPD motivates support between independent and assisted performance [Chounta et al. 2017; Cole et al. 1978; Murray and Arroyo 2002; Schulze Bernd and Chounta 2024]. We operationalize only observed reachability: a later improving submission proves that this user reached that program state, not that showing the edit causes learning or is optimal scaffolding.

Submission sequences also support knowledge tracing, which estimates latent mastery from interaction histories [Corbett and Anderson 1994; Ghosh et al. 2020; Piech et al. 2015a]. ZPDPatch instead orders source-code states using execution evidence; it neither infers mastery nor predicts a learner response. This distinction permits fully automatic repair targets while delimiting the contribution to program behavior rather than educational outcomes.

2.4 Positioning of This Work

Unlike reference alignment and LSGen, ZPDPatch does not require a same-problem solution repository; unlike prior LLM and iterative repair, its targets are mined as ordered transitions from the same user's authentic submissions. Its multi-checkpoint control must also be distinguished from deep ensembles and model soups, which aggregate predictive distributions or weights [Lakshminarayanan et al. 2017; Wortsman et al. 2022]. We instead execute complete programs, measure set-union repair coverage on validation problems, and retain separate checkpoints for conditional early stopping and structural gating. Because seed diversity alone can create complementary candidates, our evaluation gives a same-target Answer pool the identical number of trained checkpoints, exhaustive size-three choices, and test-time generation budget as the mixed-target pool.

A direct numerical port would either grant held-out tasks the method-only evidence in Table 1 or remove a defining component; query budgets also differ. We instead use matched controls: Zero-shot for adaptation, Answer for target/checkpoint diversity, and LSGen for same-problem retrieval under the common runner.

3 Problem Formulation

For problem p, let a student's i-th submission be $s_i = (c_i, v_i)$, where c_i is the complete source code and v_i is the online-judge verdict. A submission trajectory $\tau = [s_1, s_2, \ldots, s_n]$ is chronologically ordered and terminates at the first accepted submission. Let d_p contain the problem statement and resource constraints, and let $\mathcal{E}_p$ be the available test suite. Re-executing c_i gives a per-test outcome vector $o_i = (o_i(e))_{e \in \mathcal{E}_p}$.

A repair example consists of problem context d_p, an observed sequence h, and a complete target program y. Adapter-specific rules decide which submissions remain in h and which later submission supplies y; the two need not be adjacent in the original trajectory. Given $x = (d_p, h)$, a model generates one complete program $\hat{c}$.

We call a later state *reachable* when it occurs in the same student's observed trajectory and satisfies an adapter-specific improvement rule. Reachability is an empirical property of the data, not a guarantee that the transition is minimal, comprehensible, or caused by instructional support. Our design hypothesis is that learning from such transitions can produce repairs that preserve more of the current solution than direct answer generation.

4 Approach

4.1 Design Requirements

Four requirements shape the method: *automatic supervision* recoverable from submissions and tests; *target diversity* spanning intermediate improvement and accepted solutions; *executable validation* independent of model confidence; and *non-regression* when no candidate improves observed behavior. We separate these offline and online decisions. Label relations determine what each adapter learns; execution and selection determine what the system may return. A noisy target can influence generation without automatically overriding a better current program.

4.2 Trajectory Construction

We group submissions by problem and student, sort by time, and stop at first acceptance. After indentation/comment/blank-line normalization, we remove only consecutive duplicates; non-consecutive revisits remain evidence of distinct attempts. Complete source, verdict, and order are preserved, with no maximum history or transition count. Every state is re-executed. Cache keys bind problem, normalized code, test-suite identity, timeout, and memory limit, preventing incompatible reuse. Runner failures receive an explicit most-severe outcome, and Progress construction requires a complete cache.

4.3 Adapter-Specific Supervision

We order verdict severity as $\mathrm{sev}(\texttt{AC}) = 0$, $\mathrm{sev}(\texttt{WA}) = 1$, $\mathrm{sev}(\texttt{TLE/MLE}) = 2$, $\mathrm{sev}(\texttt{RE}) = 3$, $\mathrm{sev}(\texttt{CE}) = 4$, and 5 for internal failures. This is an explicit operational total preorder for mining labels, not a claim that, for example, every WA program is semantically closer to correctness than every RE program. For equal, non-empty test keys, $o_t \preceq_{\mathrm{sev}} o_r$ iff no test outcome in o_t is more severe than its counterpart in o_r. Test-case progress is the Pareto condition

$$\mathrm{TCProg}(r, t) = \mathbf{1}[o_t \preceq_{\mathrm{sev}} o_r \wedge o_t \neq o_r]\,. \tag{1}$$

Table 1. Functional comparison with execution-aware educational and neural APR. "Adaptive" means that a generated candidate changes a later model query.

System	Method-specific evidence	Query/control role
SelfAPR [Ye et al. 2022]	Diagnostic; project-local synthetic data	Static; diagnostic-conditioned generation
PyDex [Zhang et al. 2024]	Statement, tests, selected few-shots	Adaptive; tests select iterative candidates
CREF [Yang et al. 2024]	Tutor hint, solution, failing tests	Adaptive; tutor conversation
CEGIS repair [Orvalho et al. 2025]	Tests, reference, MaxSAT sketch	Adaptive; counterexample refinement
ZPDPatch	Statement, tests, trajectory; no repair-time reference	Non-adaptive generation; execution-conditioned stop, fallback, and TED gate

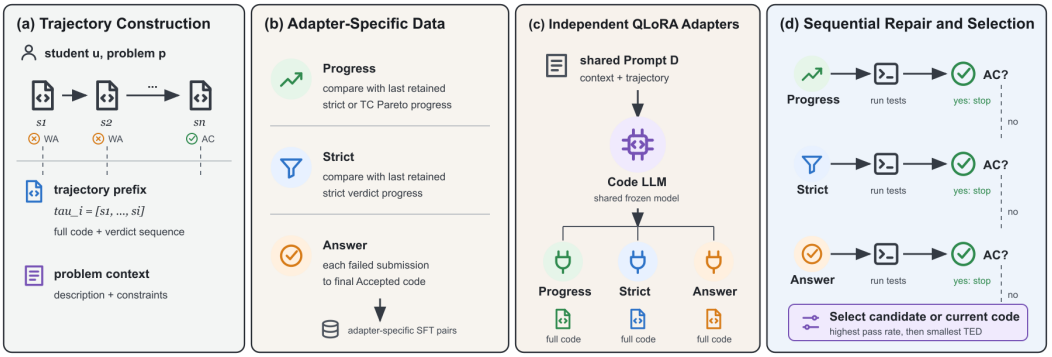

Fig. 1. Overview of trajectory supervision and execution control. The displayed P–S–A route is the one-per-relation control; the deployed system trains three seeds per relation and uses validation execution to select any three checkpoints, so a relation may repeat. Execution outcomes determine early stopping and final selection.

Answer. If s_n is accepted, every preceding non-accepted submission is independently paired with c_n. For $\tau = [S_1, \ldots, S_9]$, the rule produces $([S_1], c_9), \ldots, ([S_8], c_9)$. Thus, ANSWER learns a direct failed-program-to-answer mapping and receives no multi-submission history.

Strict. Starting with $R = [s_1]$, we scan the trajectory and append s_t only when its verdict is strictly better than the last retained submission $s_r = R[-1]$: $\mathrm{sev}(v_t) < \mathrm{sev}(v_r)$. If $R = [r_1, \ldots, r_m]$, we create $([r_1, \ldots, r_{k-1}], c_{r_k})$ for every $k = 2, \ldots, m$. The comparison is against the last retained state, not necessarily the immediately preceding original submission.

Progress. PROGRESS uses the same retained scan but appends s_t when either its verdict strictly improves or $v_t = v_r$ and $\mathrm{TCProg}(r, t) = 1$. It therefore captures monotonic per-test progress that is hidden by an unchanged coarse verdict. Prefix-target construction is otherwise identical to STRICT.

The datasets are constructed independently and may overlap. A strict transition is also a progress transition, and final accepted transitions can occur in both. The objectives nevertheless differ: ANSWER learns direct completion from one failed program, whereas STRICT and PROGRESS learn transitions from retained prefixes.

The order is consequential and auditable. Of the retained training labels, 89.5% of Strict and 90.7% of Progress remain valid if RE is placed below WA; 82.6% and 84.6% remain valid when every non-AC verdict is tied. This retrospective label audit does not estimate retrained-model performance, so we treat the chosen order as part of the method specification rather than a universal correctness ranking.

4.4 Execution-Evidence Order

The three rules are not unrelated task labels. They form two nested execution-evidence orders plus a terminal closure. Let the evidence of submission (s) be (e(s)=(v(s),o(s))) over the fixed testcase universe $\mathcal{E}_p$. Define the strict relation

$$s \prec_S t \iff \mathrm{sev}(v(t)) < \mathrm{sev}(v(s)), \tag{2}$$

and the progress relation

$$s \prec_P t \iff s \prec_S t \ \vee \ (v(s) = v(t) \wedge o(t) \prec_\times o(s)), \tag{3}$$

where $o(t) \prec_\times o(s)$ is the strict product order: every testcase outcome is no worse and at least one is better under severity. Strict and Progress retain a chain under $\prec_S$ and $\prec_P$, respectively. Answer is the terminal closure $A(s) = s_n$ from each failed state to the first accepted state; it is intentionally not an intermediate order.

Strict is a coarse projection, Progress its non-compensatory testcase refinement, and Answer a closure to acceptance. For T tests, every transition decreases $\rho(s) = (\mathrm{sev}(v(s)), \sum_e \mathrm{sev}(o(s,e)))$ lexicographically, so chains are acyclic and have length at most $6(5T+1)$. Moreover, $\prec_S \subseteq \prec_P$; these facts induce the fine-to-broad controller order.

Unlike a passed-test scalar, the product order cannot trade a newly broken test for another repaired test. Its guarantee is limited to supervision construction: learned candidates still require execution and the original fallback. The order is defined over one fixed testcase universe and severity map; cache keys bind both tests and resource limits, and Progress fails closed on incomplete vectors.

4.5 Prompt and Adapter Training

All adapters share one role-neutral prompt renderer: statement and limits, followed by every submission stored in that training record in chronological order, with verdict/outcome/edit summaries, and a request for one complete solution in the dataset's language and format. The construction rule therefore determines the training context: Answer records contain only the current failed submission, whereas Strict and Progress records contain the retained prefix through that submission. No instruction names a policy or asks for a particular edit size, so specialization can arise only from the input–target distribution. Only assistant target tokens contribute to loss.

Each adapter independently minimizes mean target-token cross entropy with the base weights frozen and its own QLoRA parameters; prompt tokens are masked.

4.6 Validation-Selected Execution Portfolio

Training loss ranks checkpoints within one target distribution, but it does not measure the cross-policy repair complementarity used at deployment. We therefore select the portfolio using execution on validation problems. Let $V = V_P \dot\cup V_S \dot\cup V_A$ be independently trained Progress, Strict, and Answer checkpoints and let $R_v \subseteq U$ be the validation instances repaired by checkpoint v. For a portfolio S, define execution coverage

$$f(S) = |\bigcup_{v \in S} R_v| / |U|. \tag{4}$$

The deployed search family is $\mathcal{F}_3 = \{S \subseteq V : |S| = 3\}$. It allows validation evidence to choose both target relations and training seeds rather than assuming that one checkpoint per relation is optimal. We retain $\mathcal{F}_{\mathrm{rel}} = \{S \in \mathcal{F}_3 : |S \cap V_a| = 1 \ \forall a\}$ as the direct relation-heterogeneity control. We sample one eligible trajectory per validation problem by a fixed hash before any execution, yielding 461 problems with complete nine-checkpoint evaluation and preventing high-volume assignments from dominating selection. In addition to unrestricted f, let R_v^B contain repairs whose AST TED is at most B, and predeclare $\mathcal{B} = \{5, 10, 20, 40, 80, 160\}$. Define $f_B(S) = |\bigcup_{v \in S} R_v^B| / |U|$.

The budget-aware objective $f_{\mathcal{B}}(S) = |\mathcal{B}|^{-1} \sum_{B \in \mathcal{B}} f_B(S)$ averages the complete coverage curve for deployments that require one budget-agnostic triple. When the budget is declared before a query, the aligned controller instead freezes the validation maximizer for that budget and invokes its triple under the corresponding gate. We exactly enumerate all $\binom{9}{3} = 84$ members of $\mathcal{F}_3$ and the 27 relation-constrained triples for unrestricted, budget-agnostic, and each budget-indexed objective. The three Answer checkpoints form the prespecified same-target control. Final test-split outcomes never enter selection.

The objectives f, f_B, and their mean are normalized and monotone. They are also submodular [Nemhauser et al. 1978]. Enumeration returns the exact validation optimum in each declared finite family; submodularity is a property, not the algorithm used here.

Finite-family selection guarantee. For problem q, let $X_q(S) \in [0, 1]$ be portfolio coverage, $F(S) = \mathbb{E}[X_q(S)]$, and $\widehat{F}_n$ its mean over n i.i.d. validation problems. Let an empirical selector return $\widehat{S}$ satisfying $\widehat{F}_n(\widehat{S}) \geq \alpha \max_{S \in \mathcal{F}} \widehat{F}_n(S)$ over finite family $\mathcal{F}$, and let S^* maximize F. With probability at least $1 - \delta$,

$$F(S^*) - F(\widehat{S}) \leq (1 - \alpha)F(S^*) + (1 + \alpha)\sqrt{\frac{\log(2|\mathcal{F}|/\delta)}{2n}}. \tag{5}$$

Hoeffding plus a union bound controls every portfolio; inserting the α-approximation between the two deviations gives the result. Exact enumeration sets $\alpha = 1$, so regret is at most twice the uniform deviation. Its i.i.d. premise does not cover identity overlap or distribution shift; RQ6 audits both rather than treating the bound as an effect certificate.

Mechanism-identification ladder. Let $r(P)$ be held-out RR for controller P, A_1 one Answer checkpoint, A_3 the fixed three-seed Answer controller, A_9 Answer-9Choose3, and M_9 mixed-target-9Choose3. The planned contrasts

$$r(A_3) - r(A_1), \quad r(A_9) - r(A_3), \quad r(M_9) - r(A_9) \tag{6}$$

separate fixed-budget checkpoint breadth, added pool-and-validation-selection opportunity, and target composition, respectively. Only the last contrast matches pool size, 84-way search, validation instances, selector, and three-call test budget; Answer training uses more examples, making it a compute-favored control rather than an advantage for M_9. Thus a positive last contrast is evidence for mixed targets conditional on the declared seed family, whereas a null or negative result attributes coverage to seed diversity and selection.

4.7 Execution-Guided Sequential Repair

At inference, we invoke the three validation-selected checkpoints in fine-to-broad relation order, breaking same-relation ties by their frozen member order. Each checkpoint receives the same complete observed evaluation trajectory, including Answer checkpoints whose SFT records contained only one submission, and produces one complete program. This common-context inference rule holds the available evidence fixed across policies; it does not claim that Answer was history-trained. Generated candidates are never inserted into another checkpoint's prompt. This independence makes the outputs a policy portfolio rather than a synthetic trajectory whose later states may lie outside the training distribution. In unrestricted mode, an accepted candidate stops the sequence; otherwise executed candidates are compared by the certified selector below. The budgeted mode additionally continues past accepted candidates that exceed the declared structural limit. The order is fixed before evaluation and tested independently in RQ3–RQ6. We also evaluate the previously plausible alternative of appending generated execution feedback as a controlled ablation.

If all candidates fail, the selector includes the current program c_i as a fallback:

$$C = \{c_i, c^{(1)}, c^{(2)}, c^{(3)}\}. \tag{7}$$

Let $\mathrm{Pass}(c)$ be the fraction of tests passed. We define $\mathrm{ted}^{\dagger}(c_i, c) = 0$ for the unchanged fallback, the Python AST tree edit distance when both changed programs parse, and $+\infty$ otherwise. Selection is lexicographic:

$$c^* = \arg\max_{c \in C} \left(\mathrm{Pass}(c), -\mathrm{ted}^{\dagger}(c_i, c) \right). \tag{8}$$

The selector first maximizes executed correctness and then minimizes change. If no candidate improves on c_i, the unchanged fallback has TED zero and the system abstains. Sequentially generating, executing, and stopping at the first accepted candidate preserves non-interacting generation from the authentic observed trajectory; if all fail, every executed candidate remains available to Equation 8.

Certified selection and cost. Including c_i in Equation 8 makes observed-test non-regression an invariant: an early return has pass rate one, and otherwise the maximum cannot score below its fallback. With a structural budget B, the gate G_B discards changed candidates with infinite or over-budget TED, continues past an over-budget acceptance, and selects from the eligible candidates plus c_i. Thus every returned change has pass rate at least $\mathrm{Pass}(c_i)$ and $\mathrm{ted}^{\dagger}(c_i, c^*) \le B$; deployed coverage is exactly the validation set-union objective f_B. These guarantees concern observed tests and edit compliance, not hidden-test correctness or learned relational behavior.

Construction costs $O(nT)$ time and $O(n)$ storage; Answer is $O(n)$. The $A_1/A_3/A_9/M_9$ ladder trains 1/3/9/9 adapters for 5.84/17.5/52.6/37.1 RTX 5090 GPU-hours and uses 0/0/4,149/4,149 portfolio-selection validation executions. Its Seen RR is 50.1/61.7/59.8/61.5%, with 1.00/1.93/1.93/1.98 mean deployment calls. Thus A_3 dominates A_9 in unrestricted RR and cost; the nine-model pools are diagnostic and support the edit-constrained frontier, not a universally cheaper default. M_9/A_3 latency is 13.62/13.32 seconds versus 24.40 for always-three LSGen, excluding cached current execution. Nine 155MB LoRA files occupy 1.4GB; deployment loads three.

5 Experimental Design

5.1 Research Questions

RQ1: Repair Effectiveness. How effectively does ZPDPatch repair held-out trajectories from problems observed during training, compared with zero-shot and retrieval-based repair?

RQ2: Trajectory Conditioning. Does access to the full submission prefix improve repair over training and inference with only the current code?

RQ3: Adapter Specialization. Do Progress, Strict, and Answer exhibit different repair coverage and patch size?

RQ4: Sequential Escalation. Does execution-guided composition improve effectiveness without always invoking all adapters?

RQ5: Problem Generalization. Does ZPDPatch retain an advantage on problems completely excluded from training?

RQ6: Robustness and Replication. Are conclusions robust to scale, seeds, prompt and verdict choices, dependence, ordering, and the Java replications?

"Frozen" excludes final-test outcomes from training and selection. None of these analyses was externally preregistered. Each post-review split, endpoint, selector, and inference implementation was committed before final-test repair evaluation; training diagnostics selected none of them. Conformance audits are specification checks, not effect tests.

Table 2. Analysis provenance; none was externally preregistered.

Status	Analyses
Original selection-frozen	Canonical tests, mechanism ladder, RQ2, and student-heldout Java
Post-review, fixed before test evaluation	Hidden tests, 1.5B scale, exercise-heldout Java, prompt control, five-fold problem cross-fitting, and verdict-order retraining
Exploratory after canonical outcomes	Disjoint/bootstrap selection, normalized TED, retention, user strata, order replay, and effect heterogeneity

RR is the primary effectiveness endpoint. The system contrast is portfolio versus Zero-shot; the mechanism contrast is selection-matched $M_9 - A_9$. For the post-review family, pooled five-fold problem-cross-fit Seen $M_9 - A_9$ is the primary mechanism estimate and its mean over the six fixed TED budgets is secondary. Scale and Java replications repeat those endpoints; prompt and verdict-order reruns are sensitivity analyses. Individual budgets and heterogeneity strata remain descriptive and do not select the conclusion.

5.2 Dataset Construction and Splits

We combine Python submissions, statements, metadata, and test cases for problems in Project CodeNet's Python800 benchmark [Puri et al. 2021]. We retain trajectories with at least three submissions, at least one non-accepted state before the first acceptance, and a final accepted program present in the Python800 reference set. We remove post-acceptance submissions and consecutive normalized duplicates. Users must appear in at least three problems and each retained problem must contain at least two trajectories. The refined corpus contains 546 problems, 21,454 trajectories, 99,432 submissions, and 10,088 tests.

Before splitting, we tokenize every possible Answer, Strict, conservative Progress, and final-evaluation prompt-target configuration. If any configuration exceeds 4,096 tokens, we exclude the entire trajectory rather than truncating code or history. This removes 2,896 trajectories and leaves 18,558. No later `max_history`, transition-count, code, or test truncation is applied.

We sort problems by eligible trajectory count and assign the top 90% (492 problems) to *Seen* and the remaining 54 low-resource problems to *Unseen*. Within every Seen problem, we reserve at least one trajectory for Train, Validation, and Test, then obtain a deterministic 80:10:10 trajectory split. All 260 Unseen trajectories remain problem-held-out test data. Table 3 reports the resulting corpus. Users may occur in more than one split; RQ5 is problem-held-out, not user-held-out.

Adapter construction yields the training and validation sizes in Table 4. Final evaluation uses the prefix ending at the last non-accepted submission and the final accepted program as the oracle. We retain only cases for which re-execution confirms the current program as non-accepted and the oracle as accepted, leaving 997 Seen instances from 328 of the 492 Seen problems and 250 Unseen instances from all 54 held-out problems. The remaining Seen problems still contribute Train or Validation trajectories but have no Test trajectory satisfying both execution checks; this is why the abstract's test-problem count is smaller than the Seen pool.

To test the educational and language scope directly, we add TIKTOC's testcase-augmented CodeWorkout release, which contains authentic CS1 Java submissions and per-test outcomes [Duan et al. 2025]. We apply the same first-acceptance, duplicate, minimum-length, and 4,096-token whole-trajectory rules, then split by student rather than trajectory. The resulting external corpus has 1,125 trajectories from 224 students over all 17 exercises, with zero student overlap: 786/158/181 train/validation/test trajectories from 157/34/33 students. Every split contains all 17 exercises. External training contains 3,470 Answer, 1,110 Strict, and 1,386 Progress examples; validation

Table 3. Dataset statistics after whole-trajectory context filtering. "Prefix" counts possible raw prefixes before adapter-specific filtering.

Group	Role	Problems	Tests	Traj.	Sub.	Prefix	Users
Seen	Train	492	9,446	14,638	58,872	44,234	4,072
	Validation			1,830	7,283	5,453	1,283
	Test			1,830	7,068	5,238	1,291
Unseen	Test	54	642	260	965	705	237

Table 4. Adapter supervision datasets. Validation examples are excluded from training.

Adapter	Train examples	Validation examples
PROGRESS	21,416	2,685
STRICT	16,973	2,116
ANSWER	40,454	4,965

contains 730/230/276. We train the same model and hyperparameters from the same base checkpoint. Seeds 2027/2028/2029 per relation create the mixed-target pool; Answer seeds 2030–2035 complete a selection-matched nine-Answer pool. We execute both pools on the 158 student-disjoint validation states, enumerate all 84 size-three portfolios per pool plus 27 one-per-relation portfolios, and freeze every winner before testing. The 181 student-held-out final states compare M_9, A_9, and the fixed A_3; no CodeWorkout test outcome enters checkpoint or portfolio selection.

5.3 Baselines

Zero-shot. The zero-shot baseline uses the same Qwen base model and prompt as ZPDPATCH without fine-tuning. If a candidate fails, its verdict and number of passed tests are added to the conversation before the next attempt. It receives at most three generations, matching ZPDPATCH's maximum budget. Because it receives the full trajectory, comparison with zero-shot measures the complete effect of learned adapters and sequential composition, not history in isolation.

Supervised Answer ensemble. Answer-3Seed is an inference-matched learned baseline: three conventional current-code-to-accepted-answer SFT adapters use the same 40,454 examples, base model, hyperparameters, greedy decoding, and maximum three-generation budget as ZPDPATCH, differing only in training seed. Answer-9Choose3 is the selection-matched control: nine independently seeded Answer adapters receive the identical exhaustive $\binom{9}{3} = 84$-portfolio validation search as the nine-checkpoint mixed-target pool. Their SFT records contain no earlier submissions, but at validation and test time they receive the same complete evaluation trajectory as every mixed-pool checkpoint. Thus prompt information, decoding, and portfolio opportunity are matched; the compared difference is the learned checkpoint pool induced by the training relations. Answer-9Choose3, rather than Zero-shot or Answer-3Seed, isolates whether mixed trajectory targets add coverage beyond ordinary checkpoint diversity and validation selection opportunity.

LSGen. LSGen is an iterative retrieval-based solution generator [Dai et al. 2026]. We preserve its repair-pair filtering, edit-vector retrieval, diff-guided generation, and re-retrieval, while replacing hard-coded infrastructure with our common dataset and runner. For a Seen query, the retrieval database contains buggy/correct pairs from other Seen-Train trajectories of the same problem; the same student and example are excluded. LSGen retrieves five pairs and performs at most three

iterations. We do not evaluate it on Unseen problems because exposing same-problem correct programs would violate problem holdout.

All approaches use the same base LLM, public tests, 2.5-second per-test timeout, 2GB memory limit, and a maximum of three candidate generations. Final validation-selected, external-replication, and budget-controller runs cap every new candidate at 4,096 new tokens; reused control candidates are audited against the same bound.

5.4 Implementation

We use Qwen2.5-Coder-7B-Instruct [Hui et al. 2024] and train each adapter for one epoch with 4-bit QLoRA [Dettmers et al. 2023]. The base weights use NF4 double quantization and BF16 computation. LoRA targets the q, k, v, o attention projections and gate/up/down MLP projections with rank 16, scaling 32, and dropout 0.05. Optimization uses paged 8-bit AdamW, learning rate 2×10^{-4}, cosine scheduling, 0.03 warmup ratio, and gradient checkpointing. Both canonical and CodeWorkout training use micro-batches of 2 with eight accumulation steps, for an effective batch of 16. Seeds 2027/2028/2029 repeat every target; Answer-9Choose3 adds Answer seeds 2030–2035 without changing data, hyperparameters, or greedy decoding. We select the lowest validation-loss checkpoint and run on one NVIDIA RTX 5090.

Two replications retain the same targets, selectors, and effective batch. The scale replication substitutes Qwen2.5-Coder-1.5B-Instruct and uses micro-batches of 4 with four accumulation steps. To avoid duplicating its large-vocabulary logits, the algebraically identical weighted FP32 token loss is reduced one sequence at a time with backward recomputation. The assignment-held-out CodeWorkout split uses 10/3/4 disjoint train/validation/test exercises (605/216/304 trajectories) and retrains both nine-checkpoint pools. It contains 213/150/161 students, including 151 shared train–test identities; it therefore isolates assignment transfer, whereas the student split isolates learners.

OpenAI Codex was used interactively to propose control and validation-selection design alternatives, implement dataset builders, orchestrate experiments, develop statistical-analysis scripts, and assist manuscript drafting. The authors selected the final design and acceptance criteria, executed every job on author-controlled infrastructure, reviewed code changes, and checked reported values against saved machine-readable outputs. AI-produced text was not treated as empirical evidence, and every cited work was checked against a retrievable source.

An isolated runner executes original and generated programs: Python800 uses the Python judge and CodeWorkout compiles Java 21 in a network-disabled container. We cache outcomes by problem, code, test set, timeout, and memory configuration and reuse the cache across supervision construction and evaluation. Within an evaluation split, every policy is bound to the same cached current-program verdict and pass rate; only a previously unseen generated candidate is executed, which prevents timeout-boundary noise from changing the baseline across portfolio members. The Java harness reproduces all 181 recorded accepted test programs before model evaluation.

5.5 Ablations

For RQ2, we construct adapter-specific Strict and Progress examples for both Seen and Unseen tests. *Full Trajectory* uses each retained prefix. *Current Code Only* retrains an adapter on identical examples and targets after removing every submission before the current one; its displayed position is reset to S_1 to prevent index leakage. Answer is excluded because its inputs already contain one submission. This design isolates the presence of history while holding targets, counts, model, and hyperparameters fixed.

For RQ3, each adapter independently generates one candidate for the same 997 Seen instances. For RQ4, we first compose the seed-2027 candidates with Progress–Strict–Answer early stopping

and fallback selection, then evaluate the final validation-selected size-three portfolio. Five controls separate target heterogeneity from repeated calls, prompt-state effects, training randomness, pool size, and the relation constraint. *Answer*×3 invokes the same Answer adapter up to three times with generated execution feedback; repeating the identical prompt would collapse to Answer once. *Generated Feedback* applies that update protocol to heterogeneous P–S–A. *Answer-3Seed* uses identical Answer data and targets with seeds 2027/2028/2029. *Answer-9Choose3* uses seeds 2027–2035 and the same 461-problem validation executions, exhaustive 84 triples, objectives, tie breaking, and frozen testing as the mixed-target pool. *Validation Relation-Constrained* selects one checkpoint per relation; the primary *Validation Execution Portfolio* selects any three of the nine mixed-target checkpoints. Every member receives the same complete authentic evaluation trajectory without generated feedback, even though Answer SFT used single-submission records, and all configurations share a maximum three-generation budget, execution limits, early stopping, and fallback selector. We audit matching Answer metadata and distinct adapter hashes. Effectiveness-only controls may omit AST TED when failed pass-rate ties use the earliest candidate; Answer ensembles and validation-selected portfolios use the complete production selector.

Independent-test confirmation. To separate candidate selection from success assessment, a post-review confirmatory audit partitions every Unseen problem's tests by a fixed SHA-256 hash (seed 2027) before regeneration. The observed side contains 332 tests and the hidden side 310, with at least three of each for all 54 problems. We recompute every displayed verdict, pass rate, and failure signature from observed tests only, regenerate the frozen ZPDPatch and Answer-9Choose3 members, and select candidates using only observed execution. A *joint repair* must pass both observed and untouched hidden tests; hidden outcomes never enter the prompt, generation, early stopping, tie breaking, or method selection.

5.6 Metrics and Statistical Analysis

For M instances, let c_m be the current program and $\hat{c}_m$ the selected result. $\mathrm{Pass}(c) \in [0, 1]$ is the test pass fraction. Pass Rate (PR) averages $\mathrm{Pass}(\hat{c}_m)$; Repair Rate (RR) averages $\mathbf{1}[\mathrm{Pass}(\hat{c}_m) = 1]$; Improvement Rate (IR) averages $\mathbf{1}[\mathrm{Pass}(\hat{c}_m) > \mathrm{Pass}(c_m)]$.

For successfully repaired, parseable programs, $\mathrm{TED}_{B\to F}$ measures AST distance from current to fixed code and $\mathrm{TED}_{F\to O}$ from fixed code to the recorded accepted oracle, using the standard unit-cost ordered-tree edit formulation [Zhang and Shasha 1989]. Legacy fixed-policy reports use ZSS; validation-selected portfolios and exhaustive budget replay use the equivalent APTED implementation [Pawlik and Augsten 2016], cross-checked against ZSS on identical labeled ASTs. Because node-count difference lower-bounds unit-cost TED, pairs whose difference already exceeds the largest declared budget are stored only as > 160, which preserves every budget decision. RQ2 uses $\mathrm{TED}_{F\to T}$, where T is the identical adapter-specific recorded target shared by Full and Current configurations.

We report Wilson 95% intervals for RR and IR [Wilson 1927]. Paired RR comparisons use two-sided exact McNemar tests [McNemar 1947], with Holm correction within each planned comparison family [Holm 1979]. Because trajectories from one problem are dependent, our primary robustness analysis resamples problems as clusters 10,000 times with seed 2027 and reports intervals for paired RR, PR, and IR differences. We additionally report a problem-balanced estimand that gives every problem equal weight, with a problem bootstrap interval. CodeWorkout contrasts report both exercise- and student-cluster intervals because either identity can repeat within its held-out split. Paired TED differences use 10,000 instance bootstrap resamples on jointly repaired inputs; both procedures follow the nonparametric bootstrap principle [Efron 1979]. Unless stated otherwise, percentages are absolute percentage points.

6 Results

6.1 RQ1: Repair Effectiveness

Table 6 summarizes RQ1 and RQ5. On Seen problems, the validation-selected execution portfolio repairs 613 of 997 programs (61.5%), compared with 306 (30.7%) for Zero-shot. Their Wilson RR intervals are [58.4, 64.5] and [27.9, 33.6], respectively. ZPDPatch alone repairs 342 instances, whereas Zero-shot alone repairs 35 (exact McNemar $p < 10^{-63}$). More importantly for dependent trajectories, the 30.8-point paired RR gain remains under problem-cluster bootstrap [26.4, 35.1]; the equal-problem-weight gain is 26.1 points [21.2, 30.9]. PR increases by 11.07 points [9.16, 13.02] and IR by 25.2 points [20.8, 29.4] under the same clustered analysis.

The selection-matched Answer-9Choose3 baseline is the strictest target comparison: ZPDPatch differs by +1.71 Seen RR points [−0.79, 4.29] and 0.0 Unseen points [−3.70, 3.75]. Thus unrestricted results establish portfolio repair utility, but not a general coverage gain from relation-mined targets over an equally large pool of independently trained Answer checkpoints.

The legacy LSGen controller achieves the highest Seen execution correctness: 88.0% PR, 80.2% RR, and 82.4% IR. It alone repairs 267 instances, while ZPDPatch alone repairs 80 (exact McNemar $p < 10^{-23}$); the clustered RR difference is −18.8 points [−23.0,−14.5]. The tradeoff is patch size. ZPDPatch's mean $\text{TED}_{B\to F}$ is 21.05, versus 27.25 for Zero-shot and 88.27 for LSGen. Among jointly repaired parseable instances, ZPDPatch reduces TED by 8.46 relative to Zero-shot (95% CI [5.05, 12.55]) and by 43.34 relative to LSGen (95% CI [38.47, 48.44]). RQ6 replaces the legacy early-stop replay with an always-three LSGen controller before making the final budget-frontier comparison.

Table 5 operationalizes this tradeoff rather than treating TED as evidence of pedagogical quality. Under a maximum structural edit budget, ZPDPatch has substantially higher repair coverage through $B = 80$; LSGen crosses over only at the broadest reported budget. At $B = 20$, for example, the gate returns an accepted repair for 44.0% of inputs with ZPDPatch versus 13.9% with always-three LSGen, a +30.09-point problem-clustered difference [26.45, 33.82]. Across all six predeclared budgets, the mean difference is +14.68 points [12.03, 17.43]. The comparison therefore supports a constrained deployment claim: retrieval has higher unrestricted coverage, whereas the trajectory-trained portfolio covers more inputs when large structural replacement is disallowed.

Paired outcomes show complementary success rather than marginal exchanges. On Seen, ZPDPatch and Zero-shot repair 271 jointly, with 342 versus 35 unique repairs; against LSGen the corresponding counts are 533 jointly and 80 versus 267 uniquely. On Unseen, ZPDPatch and Zero-shot repair 142 jointly, with 45 versus 13 unique repairs. Thus LSGen has higher coverage, but neither Seen success set contains the other.

PR and IR confirm that the RR gain is accompanied by partial behavioral improvement. Patch distance separates repair from replacement: mean current-to-fix TED is 22.8% below Zero-shot and 76.2% below legacy LSGen on each method's successful set, while the jointly repaired paired analyses above establish the same direction without success-set confounding.

6.2 RQ2: Trajectory Conditioning

Table 7 shows no consistent advantage for Full Trajectory. On Seen Strict examples, Full raises PR by 0.7 points but lowers RR and IR by 0.7 and 1.3. On Seen Progress, it lowers PR, RR, and IR by 1.0, 0.4, and 0.6. On Unseen data, Full is worse for Strict but better for Progress. Target TED is also mixed: Full is closer to the recorded target for Seen Progress and Unseen Strict, but farther in the other conditions. None of the four paired RR differences is significant after Holm correction (minimum adjusted $p = 0.313$).

Each Full–Current pair holds training examples, targets, architecture, hyperparameters, and evaluation inputs fixed; only earlier submissions are removed and position leakage reset. Direction

Table 5. Seen repair rate (%) after the same AST-TED deployment gate. ZPDPatch uses the validation-frozen budget-indexed unconstrained portfolio; LSGen always generates three candidates. The last column is ZPDPatch minus LSGen with a 95% problem-cluster interval.

Budget B	ZPDPatch	LSGen	Difference [95% CI]
5	22.5	4.8	+17.7 [14.8, 20.6]
10	32.4	8.0	+24.4 [21.2, 27.7]
20	44.0	13.9	+30.1 [26.4, 33.8]
40	50.9	30.4	+20.5 [17.0, 24.0]
80	56.0	50.8	+5.2 [1.3, 9.3]
160	57.9	67.6	−9.7 [−14.2, −5.4]

Table 6. Main repair results. PR, RR, and IR are percentages; TED is computed on successfully repaired, parseable programs.

Split	Method	PR	RR	IR	$\text{TED}_{B\to F}$	$\text{TED}_{F\to O}$
Seen	Zero-shot	76.3	30.7	43.7	27.25	27.67
	LSGen	**88.0**	**80.2**	**82.4**	88.27	83.23
	Answer-9Choose3	85.9	59.8	68.0	25.46	24.01
	ZPDPatch	87.4	61.5	68.9	**21.05**	**21.64**
Unseen	Zero-shot	75.3	62.0	68.8	17.46	15.80
	Answer-9Choose3	**85.7**	**74.8**	**80.4**	15.19	13.75
	ZPDPatch	84.9	**74.8**	79.6	**11.40**	**11.96**

Table 7. RQ2 trajectory-conditioning results. PR, RR, and IR are percentages. Current and Full use identical examples and targets.

Split	Adapter	Input	N	PR	RR	IR	$\text{TED}_{F\to T}$
Seen	Strict	Current	2,151	56.8	31.2	54.6	40.23
	Strict	Full	2,151	57.5	30.5	53.3	41.05
	Progress	Current	2,674	58.4	26.3	49.7	36.23
	Progress	Full	2,674	57.4	25.9	49.1	33.15
Unseen	Strict	Current	322	66.7	56.2	71.4	20.20
	Strict	Full	322	65.4	53.4	70.8	19.27
	Progress	Current	378	65.7	52.1	66.4	17.63
	Progress	Full	378	66.5	53.7	68.3	18.16

changes across conditions, clustered RR intervals include zero (Unseen Strict reaches zero at its upper endpoint), and target TED is mixed. RQ1 can therefore establish complete-system utility while RQ2 does not support history serialization as a stable causal explanation.

We also searched directly for cases in which history should be most identifiable. A mechanically specified matched audit requires different users on the same problem, identical current verdict and pass rate, at least one earlier submission each, and normalized current-code similarity of at least 0.8. Only 6 Strict and 12 Progress Seen examples qualify (none Unseen); Full reaches 16.7/58.3% RR versus 66.7/66.7% for Current. These tiny, post-hoc subsets cannot estimate a population effect, but

687 they expose rather than conceal that the benchmark contains little counterfactual support for a
688 history-sensitive claim.
689
690 ### 6.3 RQ3: Adapter Specialization
691 Table 8 exposes a coverage-size tradeoff. Answer has the highest PR, RR, and IR, but also the
692 largest mean patch at TED 20.22. Progress has lower coverage but the smallest patch at TED 15.85.
693 Strict and Progress have statistically indistinguishable RR (adjusted $p = 0.857$). Answer repairs
694 5.2 points more than Progress and 5.5 points more than Strict; problem-cluster intervals are [2.5,
695 7.9] and [2.9, 8.0], respectively.
696 The raw TED ordering could reflect different repaired subsets, so we also pair joint repairs and
697 resample problems. On 350 Answer–Progress joint repairs, Progress is 1.84 TED smaller [0.50, 3.26];
698 on 361 Strict–Progress joint repairs, it is 1.67 smaller [0.53, 2.93]. Strict differs from Answer by only
699 0.29 [−1.09, 1.64] across 358 joint repairs. Thus the supported specialization is Progress's narrower
700 patch scope, not a fully ordered three-policy hierarchy. Two source-based measures corroborate
701 rather than merely rename TED. On joint Progress–Answer repairs, Progress retains 1.29 more
702 percentage points of buggy tokens [0.02, 2.57] and 2.89 more points of non-empty source lines [1.16,
703 4.71]. Progress also retains 0.84 more token points than Strict [0.04, 1.67], while the line interval
704 crosses zero. These are preservation measures, not learner outcomes; hard non-regression enters
705 only through selection.
706
707
708 ### 6.4 RQ4: Sequential Escalation
709 Independent-policy escalation reaches 59.5% RR, 9.4 points above the strongest individual adapter,
710 Answer; its problem-cluster interval [7.6, 11.3] excludes zero. Gains over Progress and Strict are
711 14.6 [12.3, 17.0] and 14.9 [12.8, 17.1] points, respectively. The increase appears in PR and IR as well,
712 so the portfolio is not trading complete repairs for weaker partial outputs.
713 Progress accepts 447 inputs, Strict accepts 60 more, and 490 reach Answer. The resulting 2,037
714 generations average 2.04 per input, avoiding 954 (31.9%) of an always-three budget. Final selection
715 returns the unchanged program 329 times, making certified abstention a substantial behavior rather
716 than a corner case.
717
718 *Answer to RQ4.* The complete ladder separates four mechanisms. Independent training diversity
719 is the dominant unrestricted-coverage effect: $A_3 - A_1 = +11.63$ points with a problem-cluster
720 interval [9.54, 13.74]. Enlarging only the Answer pool does not improve it: $A_9 - A_3 = -1.91$ [−4.30,
721 0.32]. With pool size and validation selection matched, $M_9 - A_9 = +1.71$ [−0.79, 4.29] under the
722 instance-weighted estimand; the equal-problem contrast is +3.29 [0.29, 6.43]. Thus mixed targets
723 do not establish a robust unrestricted RR advantage. Repeating one checkpoint with generated
724 feedback reaches only 55.2%, and compulsory one-per-relation selection reduces RR to 57.3%.
725 Relation targets instead produce a constrained-repair effect. Across six predeclared TED budgets,
726 budget-specific M_9 portfolios improve mean Seen RR over selection-matched A_9 by 1.22 points
727 [0.02, 2.42], with TED 20 at +1.61 [0.39, 2.85]. On 532 jointly repaired inputs, M_9 preserves 2.57
728 more buggy-token points [1.56, 3.64] and 3.33 more non-empty-line points [1.89, 4.83]. The method
729 contribution is therefore target-dependent patch scope under an executable deployment constraint,
730 while checkpoint diversity explains most unrestricted coverage.
731 For auditability, using relation initial plus the last two seed digits, the validation-frozen M_9/A_9
732 triples at $B = 5, 10, 20, 40, 80, 160$ are, respectively: A29–P27–S27/A30–A33–A35, A27–P29–S27/A28–
733 A33–A34, A27–A29–P27/A27–A29–A33, A27–A28–P29/A27–A31–A33, A27–A28–P27/A28–A32–
734 A33, and A27–A28–P27/A32–A33–A34. Test outcomes did not choose any member.
735

Table 8. RQ3 individual-adapter results on 997 Seen instances.

Adapter	PR	RR	IR	$\text{TED}_{B\to F}$	$\text{TED}_{F\to O}$
Progress	73.4	44.8	52.0	**15.85**	**15.89**
Strict	74.8	44.5	51.2	17.45	18.38
Answer	**77.5**	**50.1**	**57.5**	20.22	20.61

Table 9. RQ4 mechanism ladder on 997 Seen instances. A_k uses only Answer targets; M_9 selects from the nine mixed-target checkpoints. Both size-nine pools use the same validation executions, 84 triples, and three-call budget.

Configuration	PR	RR	IR
A_1: one Answer checkpoint	77.5	50.1	57.5
Answer×3	84.1	55.2	62.4
Fixed P–S–A	86.5	59.5	67.0
A_3: fixed Answer seeds	87.0	**61.7**	**69.1**
A_9: Answer-9Choose3	85.9	59.8	68.0
M_9: mixed-target Choose3	**87.4**	61.5	68.9
Relation-constrained M_9	85.8	57.3	65.4

A post-hoc size-normalized audit applies the same gate to the validation-frozen unrestricted triples on 966 parseable-current inputs. At median 110-node programs (IQR 74–190), absolute TED 5/10/20/40 represents 4.5/9.1/18.2/36.4% of the current AST; TED 5/10 is at most 10% of the AST for 91.9/56.8% of inputs. Thus 5–10 operationalizes broadly local edits, whereas the larger gates deliberately admit substantial rewrites. At TED/current-AST-node limits 0.10, 0.20, and 0.40, $M_9 - A_9$ is +2.80 [1.04, 4.61], +3.83 [1.81, 5.95], and +2.90 [0.61, 5.23] points; limits 0.05, 0.80, and 1.60 include zero. This robustness result reduces size confounding but neither replaces the predeclared absolute frontier nor licenses a learner-quality claim.

6.5 RQ5: Problem Generalization

On 250 Unseen instances, ZPDPatch obtains 84.9% PR, 74.8% RR, and 79.6% IR, versus 75.3%, 62.0%, and 68.8% for Zero-shot. ZPDPatch alone repairs 45 cases and Zero-shot alone repairs 13 (McNemar $p = 3.01 \times 10^{-5}$). Problem-cluster intervals exclude zero for RR (+12.8 points [6.83, 18.75]), PR (+9.56 [5.66, 13.64]), and IR (+10.8 [4.76, 16.95]); the equal-problem-weight RR gain is 12.38 points [6.01, 18.62]. ZPDPatch also reduces mean successful-patch TED from 17.46 to 11.40. Among 122 jointly repaired parseable instances, its patch is 6.41 TED units smaller on average (95% CI [3.43, 9.89]).

On Unseen, Independent Policies exceeds repeated Answer-with-feedback, 73.2% versus 68.4% RR (15 versus three exclusive repairs; McNemar $p = .00754$), but matches independent Answer-3Seed: 73.2%, difference 0.0 [−3.2, 3.1], $p = 1.0$. Thus problem-held-out evidence supports checkpoint diversity, not relation heterogeneity.

The frozen unrestricted M_9 and selection-matched A_9 winners both reach 74.8% Unseen RR. Their paired difference is 0.0 points [−3.70, 3.75], so enlarging the candidate pool does not reveal a mixed-target coverage effect. Budget-indexed $M_9 - A_9$ averages +1.40 points [−1.24, 4.02] across the declared frontier; unlike the Seen contrast, this interval also includes zero. The problem-held-out evidence therefore supports portfolio coverage over zero-shot, not relation-specific coverage or a generalized constrained advantage.

Answer to RQ5. The 45 versus 13 discordant pairs establish ZPDPatch's paired advantage over Zero-shot without same-problem training. Because trajectory availability, not randomized difficulty, defines Unseen, its higher absolute RR does not imply that novelty helps repair.

Independent-test confirmation. With all prompt feedback and online selection recomputed from the observed partition, ZPDPatch repairs 192/250 inputs (76.8%) and 189 of those same returned patches also pass every hidden test: joint RR 75.6% [69.9, 80.5] and a 98.4% [95.5, 99.5] hidden-confirmation rate. The selection-matched A_9 control repairs 186 observed and 184 jointly, for 73.6% joint RR [67.8, 78.7] and a 98.9% [96.2, 99.7] confirmation rate. The paired counts are 13 ZPDPatch-only versus eight A_9-only joint repairs (exact McNemar $p = .383$); the equal-problem joint-RR difference is +2.55 points [−0.86, 6.39]. Therefore both methods' observed repairs almost always survive untouched tests, but this confirmation does not distinguish their coverage.

6.6 RQ6: Robustness and External Replication

Conformance, dependence, and seeds. The artifact audit finds zero construction-predicate, pass-rate-regression, or budget violations; all 687/199 selected Seen/Unseen changes parse. These are specification checks, not learned-behavior evidence. For 155 Seen instances from training-disjoint users, ZPDPatch gains 40.0 RR points over Zero-shot [30.7, 48.8], versus 26.7 [22.6, 31.0] on 842 overlap instances; the 69-instance Unseen disjoint-user estimate is +11.6 [−1.5, 25.4]; these strata do not identify memorization. An outcome-blind exploratory audit by verdict, current pass rate, history length, and fixed AST-size bins finds every $M_9 - A_9$ clustered interval includes zero, revealing no concentrated subgroup advantage. Three complete P–S–A runs give Seen RR 59.5/57.6/59.0% (SD 1.0) and Unseen RR 73.2/74.0/70.4% (SD 1.9). All six orders preserve fixed-candidate RR; P–S–A minimizes breadth (1.391) and successful-patch TED (21.10).

Selection stability and problem identity. Of 461 selection problems, 323 overlap the 328-problem Seen test. The chosen A27–A28–P27 triple is unchanged in all 461 leave-one-problem-out fits, but appears in only 44.2% of 2,000 problem-bootstrap fits. Excluding every test problem leaves 138 validation problems. It selects P27–P29–A28 from the mixed pool and A27–A32–A35 from the Answer pool. Their Seen RRs are 58.0% and 60.9%, respectively; $A_9 - M_9 = +2.91$ points [0.52, 5.34] with 91 versus 62 exclusive repairs (McNemar $p = .0233$). The full-data A_9 triple survives 98.5% of leave-one-problem-out fits but only 25.4% of problem-bootstrap fits, reinforcing that validation choice is not stable enough to attribute unrestricted coverage to relation targets.

The constrained result is less selection-sensitive but operating-point specific. Under the disjoint selectors, $M_9 - A_9$ averages +1.09 points [−0.23, 2.42] across budgets; TED 10 is +2.61 [0.86, 4.36] and TED 40 is +2.21 [0.21, 4.14], while the other four intervals include zero. The Unseen evaluation remains entirely problem-disjoint.

Student-held-out Java replication. Validation selects A29–P29–S28 under the relation constraint and A28–A29–S28 without it; the independently selected Answer pool chooses A28–A29–A31. Table 10 reports the prospective 181-trajectory CodeWorkout test. Selection-matched M_9 reaches 84.0% RR versus 83.4% for A_9: +0.55 points, with exercise/student-cluster CIs [−1.24, 2.37]/[−1.20, 2.59]. Their exclusive repairs are two versus one (exact McNemar $p = 1.0$). The relation-constrained portfolio reaches 81.8%; the matched test therefore provides no evidence that target composition changes unrestricted Java coverage.

Answer to RQ6. Problem clustering and full reruns preserve the CodeNet portfolio effect. A problem-disjoint selector reverses unrestricted target-composition rankings, while separately trained student-held-out Java models replicate high coverage but not an advantage over same-target diversity.

Table 10. Student-held-out CodeWorkout Java test (181 trajectories; percentages).

Validation-frozen portfolio	PR	RR	IR
Relation-constrained	95.5	81.8	86.2
M_9: mixed-target Choose3	96.3	**84.0**	87.8
A_9: Answer-9Choose3	**96.4**	83.4	**88.4**

7 Discussion

The history ablation is null, $A_3 - A_1$ is large, and neither $A_9 - A_3$ nor $M_9 - A_9$ is robustly positive in unrestricted RR: checkpoint diversity explains coverage, while execution controls escalation and abstention. Relation targets preserve more source and improve the Seen budget frontier, with problem-disjoint support only at particular budgets.

LSGen's same-problem programs yield higher unrestricted coverage with mean TED over four times ZPDPatch's. The gate operationalizes, rather than valorizes, edit size: ZPDPatch leads through TED 80 and LSGen when broad changes are allowed, a deployment frontier rather than a learner-outcome claim.

8 Threats to Validity

Construct validity. Tests can overfit [Qi et al. 2014; Smith et al. 2015]; the 310 hidden cases share one authored suite. TED and source retention are structural, not educational value; reachability and ZPD are not measured learner outcomes.

Internal validity. Shared execution, pairing, and seeds control alternatives, but construction and decoding choices may change results.

Data leakage and dependence. Unseen excludes training and retrieval problems, avoiding code duplication [Allamanis 2019]. Seen can share users and selection problems; clustering and disjoint-selection audits do not eliminate memorization.

External validity. The study spans Python contest code and Java CS1 code after dataset-specific retraining, but no multi-file projects. CodeWorkout and CodeNet holdouts reflect different populations and are not random difficulty samples. CodeWorkout holds out students or exercises separately, not both simultaneously.

Conclusion validity. Problem-clustered effects support the planned metrics, but TED intervals apply only to joint repairs. More architectures are needed for broader transfer.

Ethical considerations. Public code is used without re-identification; the artifact removes authorship signals and follows source-dataset terms.

9 Conclusion

ZPDPatch raises Seen/Unseen RR over zero-shot to 61.5/74.8%, but matched controls attribute unrestricted coverage to checkpoint diversity; relation targets improve edit-constrained coverage and source retention.

10 Data Availability

The anonymized replication package supplies splits, checkpoints, raw JSONL, code, tests, and pinned environments. `ARTIFACT.md` maps claims and seals counts, digests, and revision; release follows acceptance subject to dataset terms.

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